



PROPTech AND AI AUTOMATION

EstateFlow AI: From Real Estate Lead Enquiry to AI-Assisted Follow-Up

Research-led CRM with secured intake and human-reviewed AI drafting. A prototype demonstrating connected operational systems for real estate lead management.

Anil Kumar Gunda. MSc Computing. AI-Assisted Development and Automation

MARKET CONTEXT

Dubai Real Estate's Digital Transformation

Qualitative signals from DLD and general sales research motivate the prototype use case. Industry trends inform the hypothesis and do not establish measured EstateFlow commercial impact.

DLD PropTech and AI Integration

Dubai Land Department has publicly showcased an integrated model for real estate regulation and digital transformation, signalling institutional appetite for AI-enabled PropTech systems.

Source: dubailand.gov.ae, DLD London showcase

Unified AI-Powered Customer Services

DLD's unified customer service initiative connects customer channels with AI-enabled CRM services, supported by Microsoft, demonstrating real-world demand for connected lead and service management.

Source: dubailand.gov.ae, DLD and Microsoft partnership

Lead Response Research

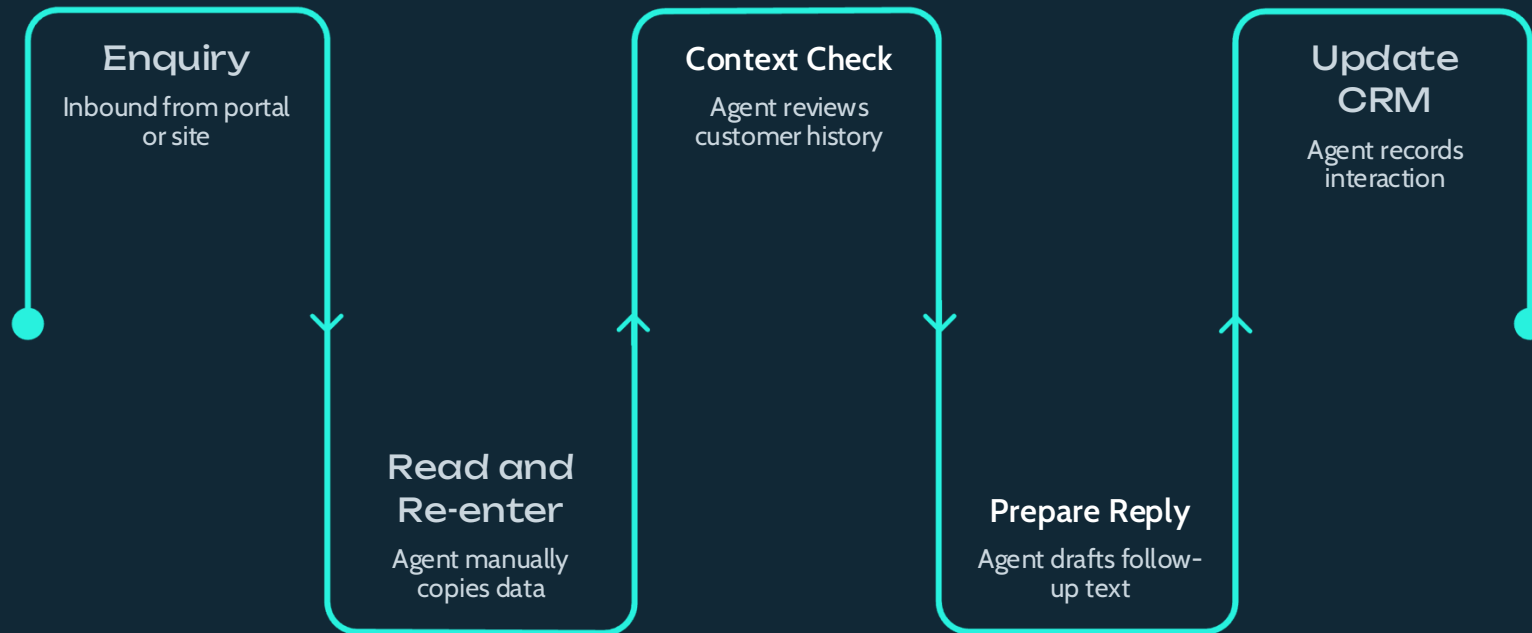
General sales research (InsideSales, 2014 Lead Response Report) motivates prompt follow-up practices. This is not a Dubai real estate specific impact statistic and should not be treated as one.

Source: resources.insidesales.com, 2014 Lead Response Report

 No numerical statistics have been asserted without a directly verified source. All figures in this deck are qualitative or source-attributed.

Hypothesized Workflow Friction

This is a selected example workflow, not a measured claim about all Dubai agencies and not deployed into a real agency. The friction points below are hypothesized and motivate the prototype.



Potential Pain Points Under Test

- Duplicate or incomplete records from manual re-entry
- Inconsistent response timing and follow-up quality
- No structured audit trail linking enquiry to agent action

Prototype Framing

EstateFlow AI is a research prototype built to test whether structured automation can address these friction points. Results reflect a controlled test environment, not a live agency deployment.

Human Oversight at the Centre

How can AI-assisted workflow automation reduce repetitive real-estate lead-management tasks while retaining structured records, traceability, and agent judgment?

AUTOMATE

Authenticated intake, webhook validation, and persistence to PostgreSQL. Structured records and activity logging on every inbound lead.

ASSIST

Lead summary and draft follow-up generated on request only. Agent explicitly triggers AI drafting from the lead detail view. Gemini does not run automatically.

HUMAN REVIEW

Agent reviews AI-generated summaries and follow-up drafts in the CRM before taking action. External communication remains a separate human action; automated message dispatch is not implemented.

📄 CRMarena (arxiv.org/abs/2411.02305) documents LLM reliability limitations in CRM tasks, reinforcing the agent-review design choice. No invented statistics cited.

Two Separate, Independent Flows

Inbound lead creation and on-demand AI drafting are **not one automatic chain**. Webhook receipt does not trigger Gemini.

Flow 1 - Inbound Lead Intake

01

External Client POST

POST `/api/webhooks/leads/` with `X-Webhook-Secret` shared-secret HTTP header

02

Constant-Time Validation

Header compared against configured `WEBHOOK_SECRET`. Wrong secret → 401, no lead created.

03

Serializer + Persistence

`LeadWebhookSerializer` validates payload → creates Lead + Activity → PostgreSQL → **201 response**

Flow 2 - On-Demand AI Drafting

01

Authenticated Agent Action

Agent opens lead detail, triggers *Generate AI Summary* or *Generate Follow-up* explicitly.

02

Django → Gemini

Django view calls `crm/ai_service.py` → Gemini API → validated response returned.

03

Save & Review

Response saved as `ai_summary` or `ai_follow_up` on Lead model. Agent reviews proposed text before any action.

Note: Shared-secret header is NOT Bearer/OAuth authentication. DRF endpoints: `/api/leads/` (CRUD/search/filter/order); `/api/leads/{id}/` (detail); `/api/leads/{id}/activities/`; AI POST actions.

What Is Verified as Built

Confirmed Components

Backend & Data

Django server-rendered HTML templates, Django authentication, Lead / Note / Activity models, Django ORM, PostgreSQL (local and managed Render production)

API & AI

Django REST Framework API, Gemini service layer (`crm/ai_service.py`), Authenticated webhook with shared-secret header

Deployment

Gunicorn and WhiteNoise on Render, GitHub repository with README, Production webhook 201 historically verified

Verified Production Events

- Login / dashboard confirmed working
- Fresh Gemini summary generated, confirmed **September 23** after Gunicorn config change
- AI follow-up production retested
- 13/13 automated tests passed, covering REST API operations, mocked Gemini integration and webhook functionality

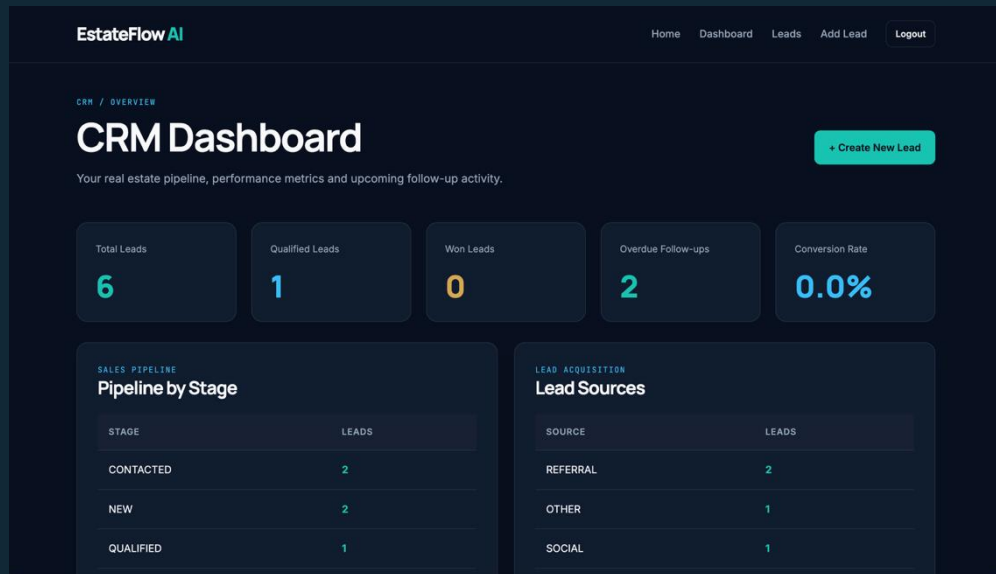
Explicitly Not Claimed

- Docker / containerized build
- Formal CI pipeline, OAuth / Bearer webhook
- n8n, dedupe, live portal integrations
- Auto-sent outbound messages, measured ROI

REAL PRODUCT OUTPUT

EstateFlow AI — Live Application Screenshots

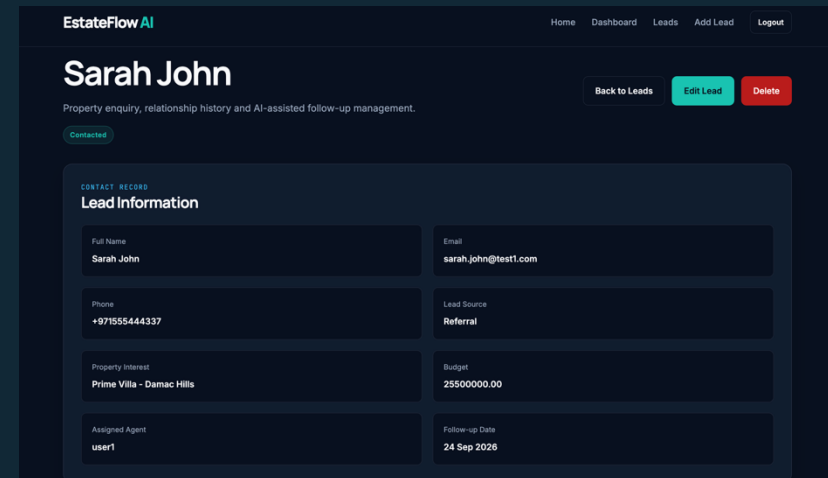
Screenshots from the deployed EstateFlow AI application, demonstrating CRM analytics, lead management and on-demand Gemini AI assistance.



The screenshot shows the 'CRM Dashboard' with a navigation bar at the top containing 'Home', 'Dashboard', 'Leads', 'Add Lead', and 'Logout'. The main heading is 'CRM Dashboard' with a '+ Create New Lead' button. Below the heading is a subtext: 'Your real estate pipeline, performance metrics and upcoming follow-up activity.' The dashboard features five key metrics: Total Leads (6), Qualified Leads (1), Won Leads (0), Overdue Follow-ups (2), and Conversion Rate (0.0%). Below these metrics are two tables: 'Pipeline by Stage' and 'Lead Sources'.

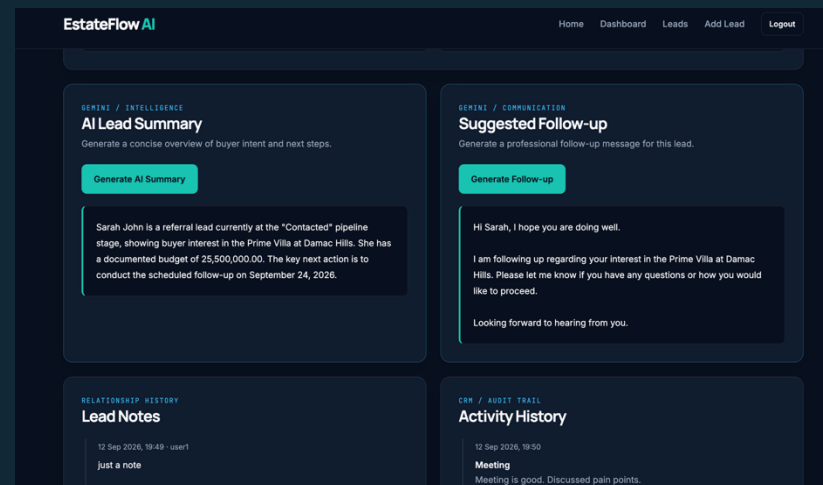
STAGE	LEADS
CONTACTED	2
NEW	2
QUALIFIED	1

SOURCE	LEADS
REFERRAL	2
OTHER	1
SOCIAL	1



The screenshot shows the 'Lead Information' page for 'Sarah John'. The navigation bar at the top includes 'Home', 'Dashboard', 'Leads', 'Add Lead', and 'Logout'. The page title is 'Sarah John' with a subtitle 'Property enquiry, relationship history and AI-assisted follow-up management.' Below the title are three buttons: 'Back to Leads', 'Edit Lead', and 'Delete'. The 'CONTACT RECORD' section displays the following information:

Full Name	Sarah John	Email	sarah.john@test1.com
Phone	+971555444337	Lead Source	Referral
Property Interest	Prime Villa - Damac Hills	Budget	25500000.00
Assigned Agent	user1	Follow-up Date	24 Sep 2026



The screenshot shows the 'AI Lead Summary' and 'Suggested Follow-up' page. The navigation bar at the top includes 'Home', 'Dashboard', 'Leads', 'Add Lead', and 'Logout'. The page is divided into four sections: 'AI Lead Summary', 'Suggested Follow-up', 'Lead Notes', and 'Activity History'.

AI Lead Summary

Generate a concise overview of buyer intent and next steps.

Generate AI Summary

Sarah John is a referral lead currently at the "Contacted" pipeline stage, showing buyer interest in the Prime Villa at Damac Hills. She has a documented budget of 25,500,000.00. The key next action is to conduct the scheduled follow-up on September 24, 2026.

Suggested Follow-up

Generate a professional follow-up message for this lead.

Generate Follow-up

Hi Sarah, I hope you are doing well.

I am following up regarding your interest in the Prime Villa at Damac Hills. Please let me know if you have any questions or how you would like to proceed.

Looking forward to hearing from you.

Lead Notes

12 Sep 2026, 19:49 - user1

Just a note

Activity History

12 Sep 2026, 19:50

Meeting

Meeting is good. Discussed pain points.

PUBLIC PROJECT COMPARISON

Design Choices Alongside Publicly Documented Real-Estate Automation Projects

RealEstateAI: github.com/neer-soni/real-estate-crm. Uptown Lead Management: github.com/ankush26/real-estate-crm

FEATURE / DESIGN CHOICE	ESTATEFLOW AI	RealEstateAI neer-soni/real-estate-crm	Uptown Lead Management ankush26/real-estate-crm
Backend framework	Django / PostgreSQL	Next.js / PostgreSQL / Prisma	Express / MongoDB
CRM data model	Lead / Note / Activity relational model	Lead, Property, Assignment, Note and Audit Log models	Lead documents with embedded notes and activities
API layer	Django REST Framework	REST endpoints for CRM and AI-agent intake	Versioned REST API (/api/v1)
AI integration	Gemini: summary and draft on request	Automated lead scoring; AI-agent integration endpoints	AI insights listed in roadmap
Webhook / API access security	Shared-secret header; constant-time compare	Auth.js and role-based access; equivalent webhook not documented	JWT and role-based access; equivalent webhook not documented
Human oversight	Agent reviews before action	Lead scoring supports agent/client workflows	Agents manage leads and update statuses
Test coverage documented	13 automated tests passed : REST API, mocked AI and webhooks.	Test count not specified in README	Automated testing listed in roadmap

TESTING

Test Scenarios : Automated & Manual Results

All 13 automated tests passed, covering REST API operations, mocked Gemini integration and webhooks. Manual verification also confirmed CRM functionality and live Gemini summary generation.

SCENARIO	EXPECTED	OBSERVED	STATUS
Login authentication	Authenticated session, dashboard access	Login and dashboard working	PASS — Manual
Create / update lead	Lead created or updated and persisted in PostgreSQL.	Lead creation and editing working	PASS — Manual
API CRUD operations	Correct status codes; data returned	Automated API tests passed	PASS — Automated
Valid webhook: 201 + lead + activity	201; lead and activity created	Production webhook returned 201; automated tests passed	PASS
Wrong shared secret: 401	401; no lead created	Webhook security tests passed	PASS — Automated
Missing required field validation	400 with field errors	API validation tests passed	PASS — Automated
Gemini summary: generate & persist	ai_summary saved on Lead	Live summary confirmed; mocked tests passed	PASS
Gemini follow-up: generate & persist	ai_follow_up saved on Lead	Mocked AI integration tests passed	PASS — Automated
Simulated provider failure	Controlled error response; no crash	Failure-handling tests passed	PASS — Automated

13/13 AUTOMATED TESTS PASSED

Production webhook creation returned HTTP 201. Live Gemini summary generation was confirmed. An earlier Unicorn SIGKILL was mitigated through configuration changes; its exact root cause was not conclusively established

What Was Built and What Comes Next

Demonstrated in This Prototype

Cloud-deployed real-estate CRM

Live on Render with authenticated agent access and relational persistence via PostgreSQL.

Secured webhook intake

Shared-secret header with constant-time comparison; 201 historically verified in production.

On-demand Gemini AI drafting

Lead summaries and follow-up drafts generated on request, saved to PostgreSQL and displayed for agent review before external action.

Documented tests and debugging

13/13 automated tests passed, covering REST API, mocked Gemini integration and webhooks. Unicorn configuration was adjusted after an earlier SIGKILL; subsequent live Gemini summary generation succeeded.

Not Yet Demonstrated

- Independently verified client time savings or conversion uplift
- Outbound messaging (WhatsApp / email dispatch)
- n8n workflow orchestration
- Duplicate handling and channel notifications
- Measured process latency benchmarks

Planned Next Steps

- n8n orchestration layer for multi-channel notifications
- Duplicate lead detection and handling
- Measured end-to-end process latency
- Role-restricted recruiter demo with isolated sample data

Links

 estateflow-ai-9utj.onrender.com

 github.com/lt077-prog/estateflow-ai

Render free-tier deployments may spin down; uptime is not guaranteed.